

Researchers wished to mimic the conditions of the medial Golgi (pH $\approx$ 6.3). Which of the buffers shown in Table 1 would be best suited for this experiment?

Table 1 Acid Dissociation Constants for Several Buffers

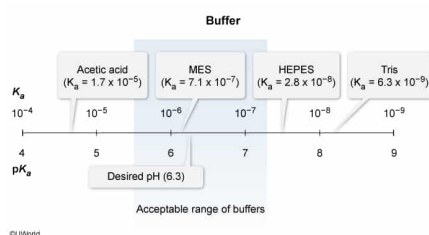
Buffer	K_a
Acetic acid	1.7×10^{-5}
MES	7.1×10^{-7}
HEPES	2.8×10^{-8}
Tris	6.3×10^{-9}

- ☐ A. Acetic acid [17%]
☒ B. MES [63%]
☐ C. HEPES [6%]
☐ D. Tris [12%]

Correct

63% answered correctly

Explanation:



A **buffer** is a mixture of a **weak acid or base** and its corresponding conjugate salt, resulting in a system that contains **both acidic and basic species** that can be protonated or deprotonated over a specific pH range. Within this range, known as the **buffering range**, the buffer can **neutralize added hydronium or hydroxide ions** and therefore **resist changes in pH**. The buffering range is generally **1 pH unit away** from the buffer's **pK_a** (the pH at which the species in the buffer are **half protonated and half deprotonated**) moving in either direction along the pH scale. Outside this range, the constituent buffer species are either fully protonated or deprotonated and the buffer loses the ability to resist pH changes. The pK_a of a buffer is related to its **acid dissociation constant K_a** by:

$$pK_a = -\log(K_a)$$

or

$$K_a = 10^{-pK_a}$$

The question states that the medial Golgi maintains a pH of approximately 6.3. To stably maintain this pH during an experiment, researchers must use a buffer with a pK_a within 1 pH unit of 6.3 in either direction (pH 5.3-7.3). A buffer

with a pK_a of 6 has a K_a of 10^{-6} , and a buffer with a pK_a of 7 has a K_a of 10^{-7} , so a buffer with a K_a value between 10^{-6} and 10^{-7} would have a pK_a near 6.3. MES meets this requirement with a K_a of 7.1×10^{-7} and would therefore be the best of the available buffers for mimicking the pH of the medial Golgi.

(Choices A, C, and D) These options have K_a values that correspond to pK_as outside the range of 5.3-7.3. They would not be good buffers at a pH of 6.3.

Educational objective:

Buffers are mixtures of a weak acid or base and its corresponding conjugate salt that can resist changes in pH by neutralizing added hydronium or hydroxide ions. They are effective within 1 pH unit of their pK_a. The pK_a of a buffer is related to its acid dissociation constant K_a by the equation $pK_a = -\log(K_a)$.